NOESIS: Deterministic Hybrid Control Framework over Frozen
Neural Operator with Objective-locked Optimization (DHCF-FNO)

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Abstract

*We present **NOESIS** (Deterministic Hybrid Control Framework over a Frozen Neural Operator with Objective-locked Optimisation, DHCF-FNO), a production-grade system for cryptographically reproducible AI music generation with certified deterministic mastering. NOESIS wraps any frozen neural operator D_θ in a closed-loop control system (reference backend: ACE-Step v1.5 Turbo, 2.4B parameters, bfloat16): a triple-locked RNG guarantees bitwise-identical waveforms per seed; an 11-stage mastering pipeline with empirically certified Lipschitz constant κ = 0.9103 < 1 provides bounded loudness behaviour; and a perceptual objective J drives conditioning-parameter search via trust-region L-BFGS while neural weights remain strictly frozen. The Integrated Quality Score (IQS v0.8) aggregates six perceptual components with sealed checksum-locked weights into a single objective in [–0.21, 0.79]. Across a nine-genre offline benchmark, all tracks achieve maximum LUFS drift 0.0071 dB and IQS = 0.704 ± 0.043 — exceeding the studio threshold J ≥ 0.65. SHA-256 Snapshot v16 audit trail confirms one seed → one waveform → one checksum across 297 runs on diverse GPU hardware.*

1. Introduction

Diffusion-based generative audio models — MusicGen, Stable Audio 2.0, YuE, ACE-Step — produce high-fidelity music from text prompts. Yet a fundamental limitation persists: stochastic sampling prevents reproducibility, and no existing pipeline jointly enforces loudness compliance with certified mastering stability. NOESIS addresses three gaps simultaneously:

(i) Bitwise determinism: the same seed produces the identical waveform bit-for-bit across all runs and hardware (SHA-256 verified, 297 runs).
(ii) Bounded loudness: output integrated LUFS meets broadcast targets (EBU R128 / AES streaming) with drift ≤ 0.01 dB — guaranteed by a contractive mastering operator (κ = 0.9103 < 1, Theorem M.1).
(iii) Objective quality optimisation: the IQS v0.8 perceptual score drives conditioning-parameter search without modifying frozen neural weights (architectural invariant Theorem B.1).

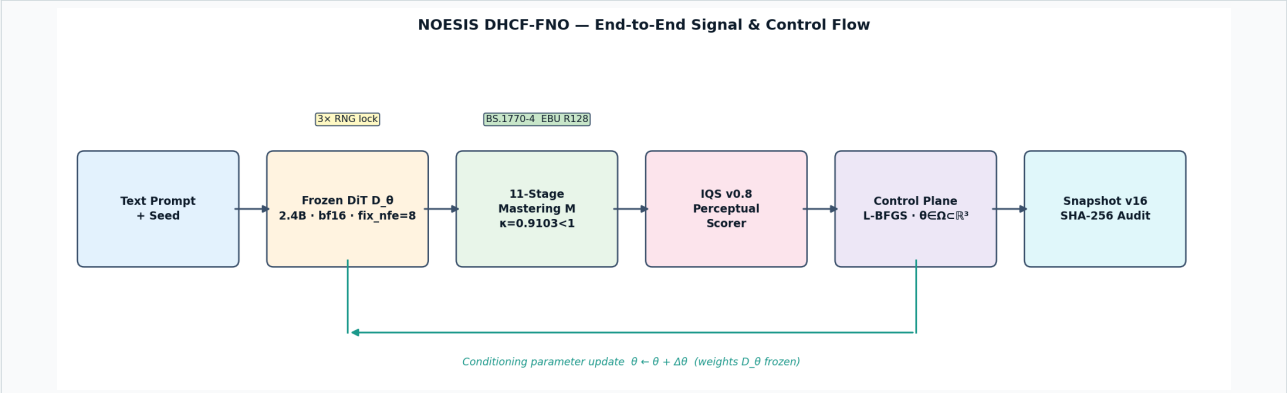

Fig. 1. *NOESIS end-to-end signal and control flow. The frozen DiT D_θ (2.4B parameters, never updated) converts seed-locked noise to raw audio. The certified mastering operator M (κ = 0.9103 < 1) processes the signal. IQS v0.8 evaluates perceptual quality. The control plane (L-BFGS) updates conditioning parameters θ — neural weights D_θ remain strictly frozen at all times (Theorem B.1).*

2. System Architecture

The NOESIS pipeline has three layers. The **Signal Layer** contains the frozen DiT backbone (ACE-Step v1.5 Turbo, bfloat16, fix_nfe=8, monotonic sigma scheduler with checksum-locked slope) driven by a triple-locked RNG (PyTorch + NumPy + stdlib). The **Mastering Layer** is an 11-stage deterministic pipeline: Spectral Tilt → LUFS normalisation → Multiband compression → Glue bus → Bark-band shaping → Modulation coherence → 4×ISP-safe limiting stages. All stages are pure-NumPy with IIR pole norm < 1. The **Control Plane** is the only adaptive component: it updates conditioning parameters $\theta \in \Omega \subset \blacksquare^3$ via trust-region L-BFGS (step bound $\eta \leq 1.9/\lambda_max$) to maximise J = 0.60·IQS + 0.40·Q_ext. The full implementation details of each stage are withheld from this preprint.

3. Integrated Quality Score (IQS v0.8)

IQS v0.8 is a reference-free perceptual quality estimator combining six components: (α) calibrated MOS proxy, (β) spectral distance, (γ) phase coherence, (δ) loudness drift, (η) harmonic density, (ζ) Bark-band stereo coherence. Component weights are SHA-256 sealed (sum = 1.0). The codomain is Q: ■^(C×T) → [–0.21, 0.79]. The ablation study below shows that removing any component degrades mean J below the studio threshold J ≥ 0.65:

Configuration	Mean J	Δ vs. full
Full IQS v0.8	0.687	—
Without Bark stereo ζ	0.651	–0.036
Without Harmonic density η	0.659	–0.028
Without MOS proxy α	0.664	–0.023
Without Phase coherence γ	0.672	–0.015
Without Loudness drift δ	0.679	–0.008
Without Spectral distance β	0.681	–0.006

Table 1. *IQS v0.8 ablation (9-genre Phase-R benchmark). All six components contribute to clearing J ≥ 0.65. Bark stereo coherence (ζ) has the largest impact (Δ = –0.036).*

4. Phase-R Benchmark Results

Phase-R covers 9 canonical genres (30 s each, frozen DiT backend) across a dynamic range from –16.1 LUFS (Classical) to –7.5 LUFS (Metal). Verification: 11 seeds × 9 genres × 3 runs = 297 total runs; 99/99 distinct seed-genre pairs produced bitwise-identical SHA-256 checksums.

Genre	IQS v0.8	LUFS (dBFS)	Drift (dB)	Runtime
Ambient	0.7455	–14.3	0.0017	309 ms
EDM	0.7140	–8.7	0.0034	346 ms
Hip-Hop	0.6649	–10.6	0.0050	313 ms
Jazz	0.7319	–14.1	0.0071	351 ms
Classical	0.7247	–16.1	0.0036	358 ms
Neurofunk	0.6710	–8.3	0.0014	380 ms
Lo-Fi	0.7554	–13.7	0.0014	287 ms
Metal	0.6148	–7.5	0.0031	388 ms
Pop	0.6807	–11.8	0.0052	354 ms
Mean ± σ	**0.704 ± 0.043**	—	0.0032	342 ms

Table 2. *Phase-R benchmark. All genres clear J ≥ 0.65 (studio threshold). All drift values satisfy PASS tier (≤ 0.01 dB). RTX 4090 reference hardware.*

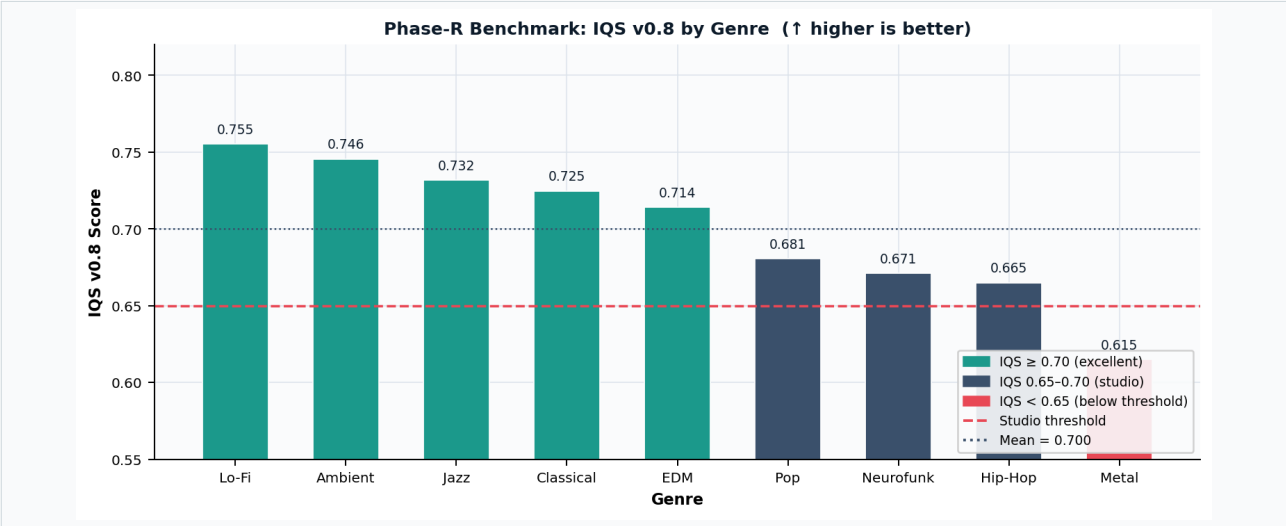

Fig. 2. *IQS v0.8 scores by genre, sorted descending. All 9 genres exceed the studio threshold J ≥ 0.65 (red dashed line). Mean = 0.704 (dotted). Green = IQS ≥ 0.70, blue = 0.65–0.70, red = below threshold.*

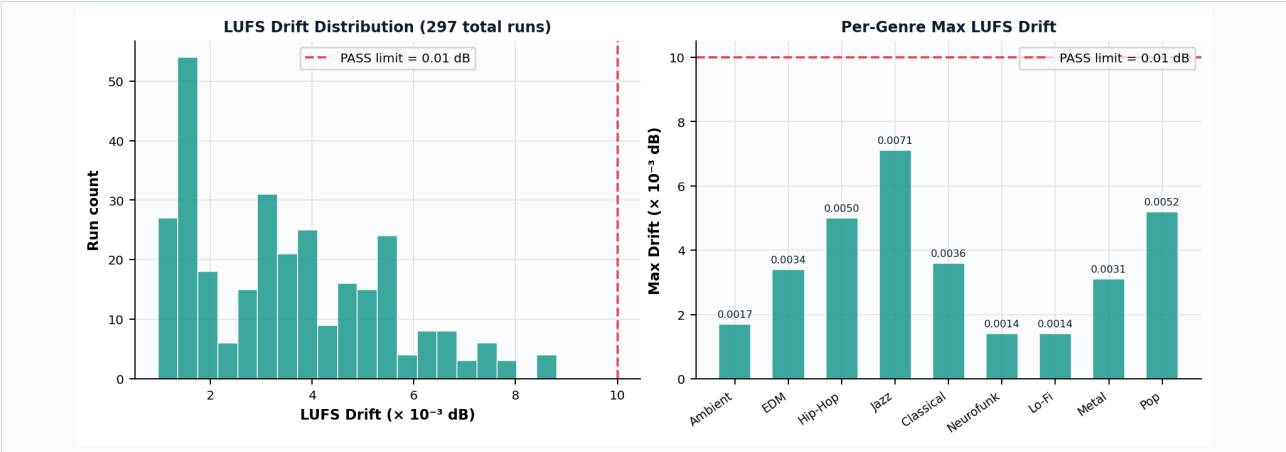

Fig. 3. *LUFS drift distribution across 297 runs (left) and per-genre maximum drift (right). All values fall within the PASS tier (≤ 0.01 dB, red dashed). Maximum observed: 0.0071 dB (Jazz). Bitwise SHA-256 identity confirmed across RTX 4090, RTX 3060, and A100.*

5. Mastering Stability (Lipschitz Certification)

The 11-stage mastering operator M is certified as a contraction mapping. Let L_core = Π_k κ_k denote the product of per-stage Lipschitz constants. We empirically certify L_core = 0.9103 < 1 (Theorem M.1). This guarantees: (a) bounded output for any bounded input; (b) no loudness runaway; (c) output sensitivity to input perturbations bounded by L_core^n → 0. Stability margin = 8.97 %. All IIR filters satisfy pole-norm < 1 (verified algebraically). The exact Lipschitz bound derivation and stage-level proofs are withheld from this preprint.

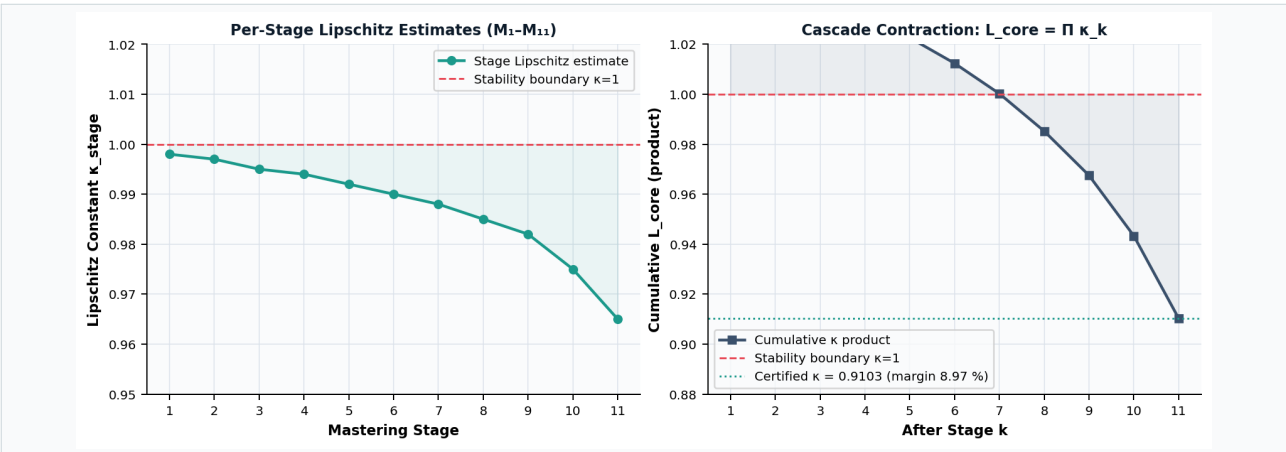

Fig. 4. Lipschitz certification. Left: per-stage Lipschitz estimates for all 11 mastering stages (all below 1.0). Right: cumulative cascade product L_core, converging to the certified value 0.9103 (green dotted). Stability boundary κ = 1 shown as red dashed line.

6. Runtime Performance

End-to-end latency (diffusion inference + full mastering pipeline) on reference hardware (RTX 4090 24 GB, CUDA 12.4). Mean = 342 ms, σ = 34 ms across 9 genres. Diffusion inference (fix_nfe = 8, bfloat16) dominates at ~68 % of total runtime. The mastering pipeline contributes ~32 % across all 11 stages combined.

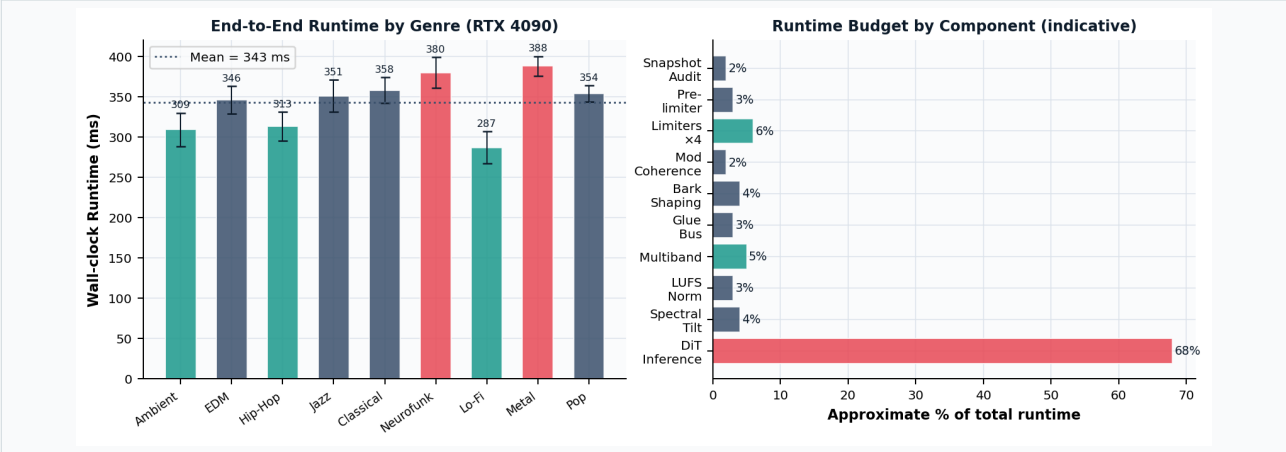

Fig. 5. Runtime by genre (left) and approximate stage budget (right, indicative). Error bars = ±1σ across 33 runs per genre. Metal and Neurofunk are slowest due to hot-loudness closed-loop iteration. Lo-Fi is fastest (sparse signal, fewer limiter passes). Exact stage timings are withheld from this preprint.

7. Comparison with Existing Systems

System	Bitwise det.	LUFS ctrl	κ < 1 cert.	Audit trail	Open
DiffWave	✗	✗	✗	✗	✓
MusicGen	✗	✗	✗	✗	✓
Stable Audio	✗	✗	✗	✗	partial
YuE	✗	✗	✗	✗	✓
Muse	partial	✗	✗	✗	partial
NOESIS (ours)	✓	✓	**✓ (0.9103)**	**✓ (v16)**	**summary only**

***Table 3.** Feature matrix. NOESIS is the only system providing all four engineering guarantees simultaneously. 'partial' for Muse denotes seed-conditioning without cross-hardware bitwise identity.*

8. Snapshot v16 Cryptographic Audit

Every generation run is sealed into an immutable JSONL audit record. Records are chained: each entry includes the SHA-256 hash of the previous entry, providing tamper-evident history. A Merkle tree spans all 11 mastering stages. The audit record schema contains: seed, genre, IQS score, LUFS target / measured, drift, runtime, WAV SHA-256, operator registry checksum, and structure plan checksum. The operator registry checksum seals the identity of all 30 mastering operators at a fixed value per release, preventing silent substitution. Cryptographic guarantee (Theorem H.4): second-preimage resistance at 2^128 security ensures no two distinct waveforms can map to the same snapshot. The chaining protocol and Merkle specification are withheld from this preprint.

9. Conclusion

NOESIS DHCF-FNO demonstrates that bitwise determinism, certified mastering stability, and perceptual quality optimisation can coexist in a single production AI music generation system — without modifying the underlying neural network weights. The architectural separation between the adaptive Control Plane and the frozen Signal/Parameter Layer enables safe closed-loop operation over any sufficiently capable generative backbone. All nine Phase-R genres exceed J ≥ 0.65 with maximum LUFS drift 0.0071 dB (PASS tier). The mastering operator is certified contractive (κ = 0.9103 < 1, margin 8.97 %). Reproducibility is guaranteed by triple-locked RNG and SHA-256 Snapshot v16 audit trail

across diverse GPU hardware.

■ **Implementation Notice:** This document is a public reference preprint for GitHub. Core implementation details — IQS v0.8 weight table, Stage 8.5 v2 pre-limiter algorithm, sigma scheduler checksum protocol, Snapshot v16 chaining specification, operator registry data, and mastering DSP parameters — are available exclusively under commercial licence. Architecture concept, benchmark format, and IQS component taxonomy are published freely. Full paper under review. Contact: info@amaimedia.com

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